

Virtual Memory

Lesson 09 — Paging, Segmentation and Swapping

JKSSB Computer Awareness

October 8, 2026

Why Virtual Memory Exists

- ▶ A program or its data can be **larger than physical RAM**.
- ▶ Virtual memory lets the system run such programs anyway, using secondary storage (disk) as an extension.
- ▶ It gives each process a **large, contiguous view** of memory, even though the real RAM is smaller and shared.
- ▶ The office desktop (EX-01) runs MS Word, a spreadsheet, and the browser together; virtual memory keeps them all addressable.

High-yield point

Virtual memory **extends** addressable memory using disk; it is **not** extra physical RAM.

Virtual vs Physical Addresses

Virtual address

- ▶ The address a program generates.
- ▶ Large, uniform space per process.
- ▶ Not the real RAM location.

Physical address

- ▶ The actual location in RAM.
- ▶ Managed by the OS.
- ▶ Where the data really sits.

Exam cue

The program always sees **virtual** addresses; the hardware converts them to **physical** addresses.

The MMU: The Translator

- ▶ **MMU = Memory Management Unit.**
- ▶ It is the hardware that **translates virtual addresses into physical addresses** at run time.
- ▶ It works with the page table the operating system maintains.
- ▶ Every memory access passes through this translation.

Trap

Translation is the MMU's job; **storing files is not**. Virtual memory uses the disk, but the MMU only maps addresses.

Paging: Fixed-Size Blocks

- ▶ Paging divides the **virtual address space into fixed-size pages**.
- ▶ Physical memory is divided into equal **frames** of the same size.
- ▶ Any page can be loaded into any free frame.
- ▶ Because all blocks are equal, allocation is simple and fast.

Exam cue

Pages are the virtual blocks; **frames** are the physical blocks. Both are fixed-size and equal.

Segmentation: Variable-Size Segments

- ▶ Segmentation divides a program into **variable-size logical segments**.
- ▶ A segment matches a program unit: code, data, stack, a procedure, or a module.
- ▶ Segment sizes differ, because logical units differ.
- ▶ It reflects the **programmer's logical view**, not a fixed grid.

Trap

Segmentation is not paging. Paging uses equal fixed-size pages and frames; segmentation uses unequal logical segments.

Paging vs Segmentation

Paging	Segmentation
Fixed-size blocks	Variable-size blocks
Blocks: pages / frames	Blocks: logical segments
Equal size	Unequal size
Invisible to programmer	Matches program structure
Simple allocation	Matches logical units

Remember: *pages are fixed; segments are logical and vary in size.*

Page Fault and Demand Paging

- ▶ A **page fault** occurs when the referenced page is **not currently in RAM**.
- ▶ The page must then be **loaded from the disk** into a free frame.
- ▶ **Demand paging** loads a page only when it is actually needed, not all at once.
- ▶ This is why a program can start before all of it is in memory.

Trap

A page fault is **not a hardware fault**. It is a normal signal that the needed page is currently on disk.

Swapping and the Disk-Backed Hierarchy

- ▶ **Swapping** moves pages or whole processes between **RAM and secondary storage (disk)**.
- ▶ Virtual memory is therefore **disk-backed**.
- ▶ The hierarchy from the CPU outwards: registers, cache, RAM, then disk-backed virtual memory.
- ▶ EX-01: with 16 GB RAM, a working set that exceeds RAM spills to disk and is loaded back on demand.

Exam cue

Disk-backed virtual memory is **much slower** than RAM. It adds **capacity**, not speed.

Not Extra RAM — and Not Cache

- ▶ Virtual memory is **not additional physical RAM**; it borrows space on the disk.
- ▶ It is **not extra speed**: reading from disk is far slower than reading from RAM.
- ▶ Do not confuse it with **cache**: cache is small, fast memory **close to the CPU**; virtual memory is **large and disk-backed**.

Common confusion

Cache *speeds up* frequently used data; virtual memory *extends capacity* using slower storage.

Thrashing

- ▶ **Thrashing** is **excessive paging**.
- ▶ The system spends most of its time moving pages between RAM and disk, leaving little time for useful work.
- ▶ It happens when too many processes compete for too little RAM.
- ▶ Performance collapses even though the CPU is busy.

Exam cue

Thrashing = **too much paging**, not a hardware failure. Adding RAM usually relieves it.

Lesson 09: Recall

- ▶ Virtual memory **extends memory using disk**; it is slower than RAM and is **not extra physical RAM**.
- ▶ **MMU** translates virtual addresses into physical addresses.
- ▶ **Paging** = fixed-size pages and frames; **segmentation** = variable-size logical segments.
- ▶ A **page fault** means the referenced page is not in RAM and must be loaded from disk; **demand paging** loads pages only when needed.
- ▶ **Thrashing** = excessive paging.